

Effects of Reading Goal and Reading Task on Eye Fixation Patterns

Jukka Hyönä, Johanna K. Kaakinen
(University of Turku)

We review two studies examining task effects on eye guidance during reading. The first study investigated effects of reading perspective on eye behavior in reading. It demonstrated that both the initial encoding of words as well as the later integration stage of wrapping up the sentence meaning are influenced by the reading perspective. Early effects of reading perspective were observed both in saccadic programming and in fixation times and were primarily seen when reading a text of familiar content. In the second study, effects of reading task were examined by comparing eye movements between proofreading and reading for comprehension. Task effects appeared very early in the processing time line; both temporal and spatial aspects of eye movements were affected. Taking together, the two studies demonstrate that readers make a global adjustment to eye behavior on the basis of reading task or goal.

We read texts for various purposes. In the mornings, many of us spend the first waking hour reading a newspaper to be up-to-date about what has been happening in the world lately. Or we may be more selective in our reading and only make effort to find out how our favorite football or basketball team has succeeded the previous night. Some of us may skip the newspaper and instead spend the early hours in evaluating the merits and demerits of a scientific article or proofread with great pleasure and pride a paper accepted for publication in a prestigious journal. Yet others may find the time appropriate for studying for an exam. In short, written texts are approached from multiple perspectives and goals.

But how does a reading goal or task affect attentional guidance during reading? Despite its potential practical and theoretical significance, surprisingly few eye movement studies have been conducted to investigate this question. On the other hand, this issue has been much more popular in educational psychology, where it has primarily been studied using self-reports (e.g., Lorch, Lorch, & Klusewitz, 1993), text recalls (e.g., Narvaez, van den Broek, & Ruiz, 1999; van den Broek, Lorch, Linderholm, & Gustafson, 2001), and think-alouds (e.g., Narvaez et al., 1999; van den Broek

et al., 2001). In the present article, we review two studies investigating how reading tasks influence eye guidance during reading. Before doing that we shortly summarize previous eye movement studies on this topic.

Previous eye movement studies on the effects of reading task

Previous studies have examined how overt or subvocal pronunciation (Heller, 1982; Hendriks, 1996), questions presented before reading (Rothkopf & Billington, 1979), a reading perspective (Kaakinen & Hyönä, 2005, 2007, 2008; Kaakinen, Hyönä, & Keenan, 2002, 2003), and reading instructions (Heller, 1982; Laycock, 1955; Radach, Huestegge, & Reilly, 2008; Shebilske & Fisher, 1983; Wotschak, 2009) affect eye movements during reading.

An early demonstration of the influence of reading instructions on fixation patterns was reported by Laycock (1955). He asked participants to read “normally” and then “as fast as possible”. Eye fixation patterns changed as a function of the reading instructions. Readers made fewer and shorter fixations and less regressions during speeded reading than during normal reading (see also Heller, 1982; Shebilske &

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Correspondence concerning this article should be addressed to Jukka Hyönä, Department of Psychology, University of Turku, FI-20014 Turku, Finland. Email: hyona@utu.fi

Fisher, 1983).

Heller (1982) examined how different vocalization instructions (silent reading, internal speaking, “murmuring”, and oral reading) influence eye fixation patterns during reading. The results showed that readers made most and longest fixations during oral reading, and least and shortest fixations during silent reading (for similar results, see Hendricks, 1996). Hendricks (1996) examined the speed of vergence eye movements (i.e., how quickly the two eyes converge after a saccade) during subvocal pronunciation and silent reading for comprehension. The vergence velocities were higher during reading for comprehension than during subvocal pronunciation. Hendricks concluded that readers selectively attend to smaller regions of text when the reading task emphasizes the sound of the words (subvocalization) than when the task emphasizes meaning (silent reading for comprehension).

Task effects on eye fixation patterns has also been studied by manipulating the type of questions asked after reading. Wotschak (2009) showed that if the reader is presented with relatively difficult questions after reading, increases in practically all fixation duration measures are observed in comparison to reading in preparation for easy questions. Radach et al. (2008) compared reading in preparation for comprehension versus word verification questions (a multiple-choice task). The word frequency effects were more pronounced in gaze durations and total viewing times (albeit the word frequency $\times$ task interactions were non-significant) when readers were expecting comprehension questions than word verification questions. Radach et al. concluded that comprehension questions encourage a deeper level of processing than the word verification questions, resulting in greater word frequency effects.

Some studies have examined the impact of reading task on how readers divide attentional resources between different parts of text. Rothkopf and Billington (1979) examined how relevance of text information to the reader’s goals influences eye fixations during the course of reading. They presented a

set of questions to the readers prior to reading and recorded eye movements for text paragraphs including question-relevant and question-irrelevant information. They found that question-relevant paragraphs attracted more and longer fixation durations than question-irrelevant paragraphs. In a related set of studies, we (Kaakinen & Hyönä, 2005, 2007, 2008; Kaakinen et al., 2002, 2003) examined how reading perspective affects eye movements during reading of longer texts. In these studies, participants were asked to adopt a specific perspective for reading the text (e.g., to imagine that they have a friend who has fallen ill with a rare disease called trigeminal neuralgia). The experimental texts were relatively long and included both perspective-relevant and perspective-irrelevant information (e.g., the text described four different diseases, one of them being trigeminal neuralgia). In all studies readers spent longer time reading perspective-relevant than perspective-irrelevant sentences—this difference was evident already during the first-pass reading of the sentences. Moreover, perspective-relevant sentences attracted longer look-backs than perspective-irrelevant sentences.

Effects of reading perspective on eye guidance during reading

As reviewed above, when readers adopt a specific perspective from which to approach a text they spend more time fixating sentences relevant than irrelevant to the adopted perspective. But what aspects of eye movements are affected by the reading perspective? Do readers make shorter saccades and skip fewer words when reading relevant than when reading irrelevant text materials? Or do they reread more words in the relevant than irrelevant sentences? It is also possible that they adjust their fixation times according to task demands.

We (Kaakinen & Hyönä, 2007) studied these questions in an experiment where we gave readers a specific perspective prior to reading. We had previously found out (Kaakinen et al., 2003) that familiarity of text contents significantly influences how readily a reading perspective may be adopted to

guide text processing. Thus, we selected two texts for the experiment, a text on familiar contents and a comparable text on unfamiliar contents. The texts were adopted from the study of Kaakinen et al. (2003). The text on familiar contents discussed the symptoms and treatment of four familiar diseases (flu, diarrhea, AIDS, and chicken pox), while the text on unfamiliar contents discussed four rare diseases (typhus, trigeminusneuralgia, scleroderma, and cystic fibrosis). Before reading the familiar diseases text, the participants were asked to imagine that they are elementary school teachers and they should prepare themselves to talk to the students about flu. They were then asked to read the text in order to be able to do the job. Before reading the unfamiliar diseases text, participants were told to imagine that their friend has fallen ill of a rare disease called trigeminusneuralgia and their job is to read the text in order to be able to tell other friends about this disease.

We expected readers to be better able to utilize an assigned reading perspective when reading a text of familiar than unfamiliar contents. Effects of reading perspective were examined by comparing eye movement patterns on a matched set of target sentences, of which 10 were relevant to the reading perspective (i.e., they said something important about the target disease) and 10 were irrelevant (i.e., they discussed aspects of a non-target disease). In other words, the examination of perspective effects was carried out using only a small subset of the text contents (the length of the entire text was 1065 words for the familiar diseases text and 878 words for the unfamiliar diseases text).

To further study possible familiarity effects in perspective taking, we asked participants to read the text twice from the same perspective¹. We predicted that readers would be able to more extensively use the adopted reading perspective in guiding their visual attention during the second reading of the text, particularly when the text was of familiar contents.

In order to obtain a more detailed picture of the

perspective-driven eye guidance, we defined three areas of interest in each target sentence: (a) sentence-initial word (i.e., two first words in the sentences), (b) sentence-final words (i.e., two final words in the sentences), and (c) sentence-medial words (i.e., all other words). If the reading perspective has an early influence on eye guidance, its effects should be observable already in the sentence beginning. On the other hand, if the reading perspective primarily affects later integrative stages of processing, the effects should be restricted to sentence-final words (i.e., the sentence wrap-up effect, Just & Carpenter, 1980; Rayner, Kambe, & Duffy, 2000).

Two measures, probability of skipping a word and probability of making a regression, were utilized to study effects of reading goal on saccadic programming. Two other measures, gaze duration and rereading time, were used to assess influences on fixation durations. For each measure, we computed an estimate of the size of the perspective effect by subtracting the mean of the perspective-irrelevant condition from the mean of the perspective-relevant condition. Thus, this difference score indicates, for example, the extent to which gaze duration on words in the perspective-relevant sentences are longer than on words in the perspective-irrelevant sentences. Note that for the skipping probability we predict the difference score to be negative (i.e., more word skipping in the irrelevant than relevant sentences).

Time course of the perspective effects during the initial reading of the text

In the following, we present results bearing on the question whether reading perspective can be more readily (i.e., faster / earlier) utilized when reading a text of familiar than unfamiliar text. These data come from the first reading of the two experimental texts. The initial encoding of words was assessed by gaze duration (i.e., the summed duration of words during their first-pass reading) and skipping probability (i.e., probability of skipping over a word dur-

1. After second reading, we asked participants to switch perspective to another disease (i.e., to the matched irrelevant disease) and read the text once more from this new perspective. We were interested in how much of the originally irrelevant (i.e., during the first two readings) text contents were processed to help process the same content as perspective-relevant (i.e., during the third reading). However, these data are not discussed here.

ing the first-pass reading). Later integrative processing was assessed by rereading time and probability of regression. As noted above, the earliest effects were indexed by the two initial encoding measures for the sentence-initial words.

Measures of initial encoding. The size of the perspective effect in the two measures of initial encoding is presented in Figure 1 and 2. In *gaze duration* (Figure 1), the perspective effect showed up most strongly at the sentence end, as a sentence wrap-up

effect. However, the effect was also significant (although smaller) for the sentence-initial and sentence-medial words. Words in the familiar diseases text were read with shorter gaze durations (not apparent in Figure 1) than words in the unfamiliar diseases text. However, the perspective effect was of similar size for the low prior knowledge (unfamiliar) and high prior knowledge (familiar) texts. Moreover, the perspective effect did not show up earlier for the familiar than the unfamiliar text.

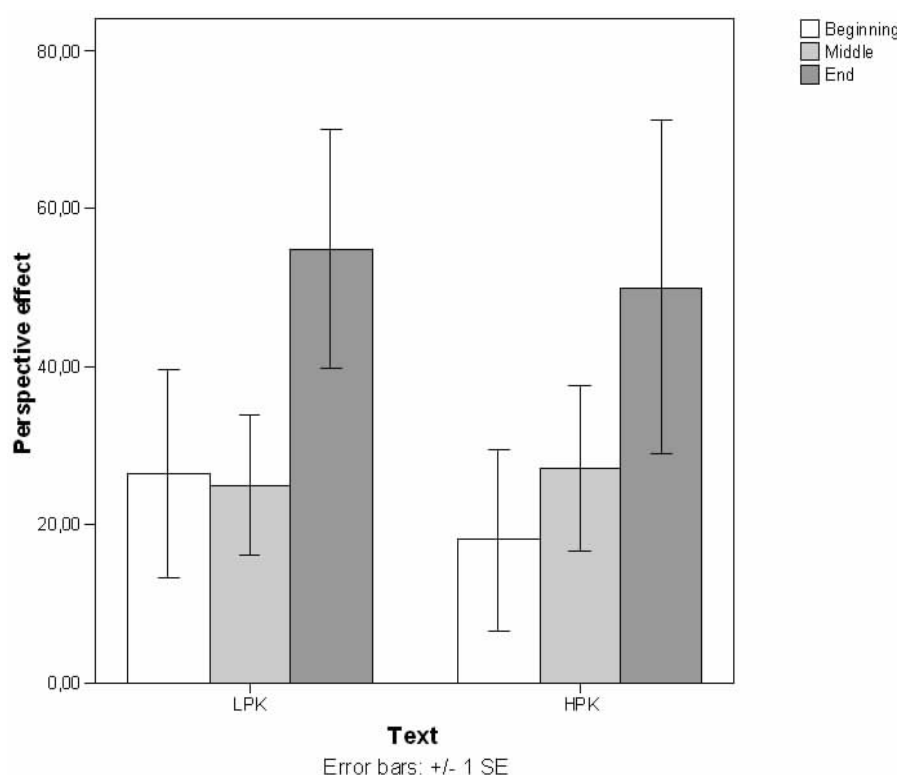


Figure 1. Size of the perspective effect (perspective-relevant – perspective-irrelevant) in gaze duration, as a function of sentence position (beginning, middle, end) and familiarity of text contents (HPK = high prior knowledge text; LPK = low prior knowledge text).

The probability of skipping (see Figure 2) demonstrated a reliable perspective effect only for the text of familiar contents. Note that the direction of the effect is negative indicating that readers skipped more words in the perspective-irrelevant than perspective-relevant sentences. In all, the skipping data provide evidence that saccadic programming can be modulated by the reading goal when reading materials readers have ample prior knowledge of. The data also

showed (not apparent in Figure 2) that words in the text of familiar contents were generally skipped more than words in the unfamiliar text.

Measures of integrative processing. Figures 3 and 4 present the size of the perspective effect in two eye movement measures, probability of regression and rereading time, indexing integrative processing. The probability of regression measure indicates when during the target sentence reading a regression to-

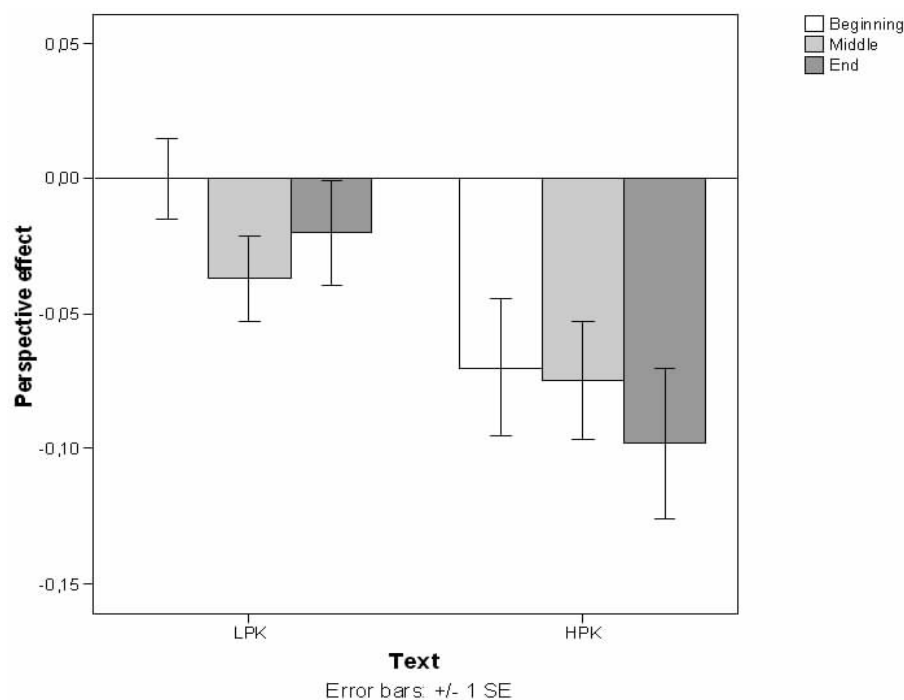


Figure 2. Size of the perspective effect (perspective-relevant - perspective-irrelevant) in the probability of skipping words, as a function of sentence position (beginning, middle, end) and familiarity of text contents (HPK = high prior knowledge text; LPK = low prior knowledge text).

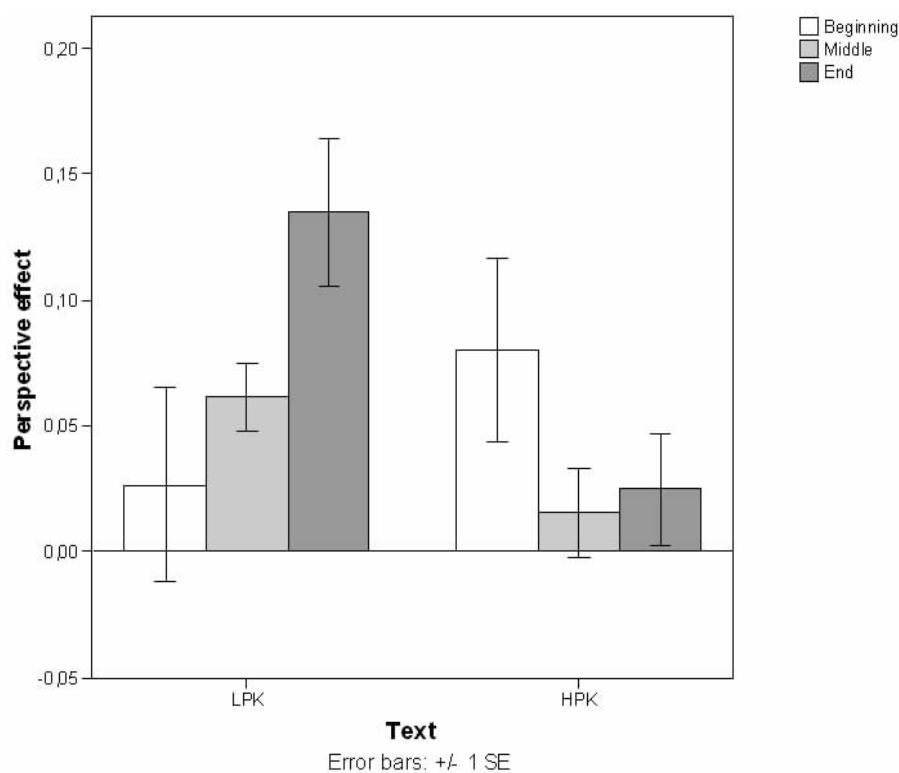


Figure 3. Size of the perspective effect (perspective-relevant - perspective-irrelevant) in the probability of making a regression, as a function of sentence position (beginning, middle, end) and familiarity of text contents (HPK = high prior knowledge text; LPK = low prior knowledge text).

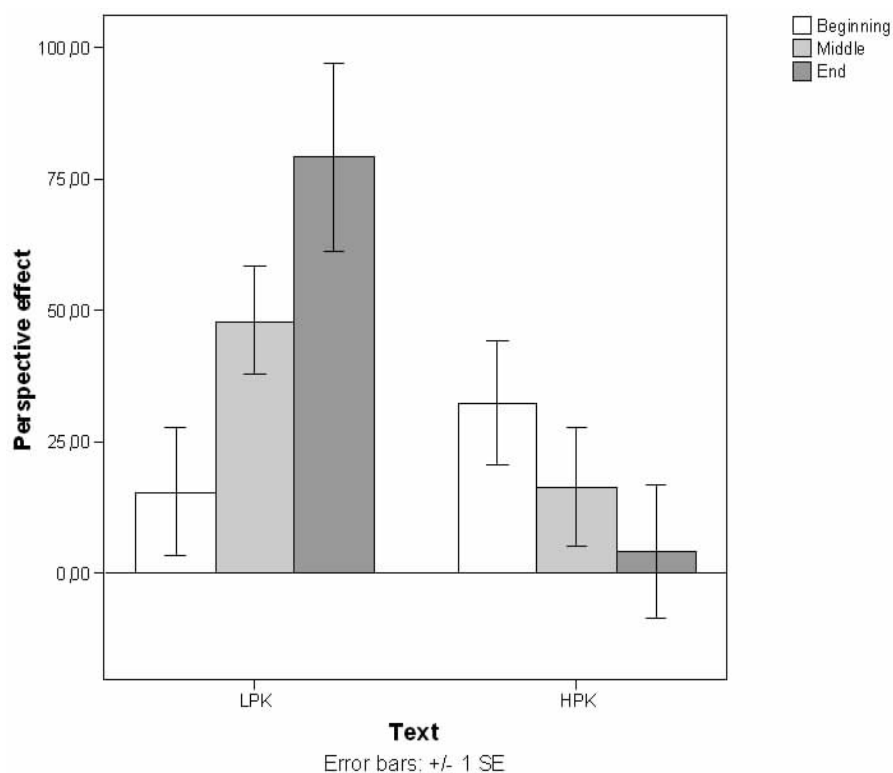


Figure 4. Size of the perspective effect (perspective-relevant – perspective-irrelevant) in rereading time, as a function of sentence position (beginning, middle, end) and familiarity of text contents (HPK = high prior knowledge text; LPK = low prior knowledge text).

ward an earlier part of the sentence is launched. As is evident from Figures 3 and 4, the perspective effect in both regression probability and rereading time was modulated by prior knowledge and sentence position. As the two measures yield a highly similar patterns of results (as they should), we only discuss here the data on the regression probability.

In the unfamiliar text, the perspective effect in the probability of regression showed up primarily at the sentence end (it was also significant in the sentence middle, but not in the sentence beginning). In contrast, in the familiar text the perspective effects were manifested in the sentence beginning. Thus, readers are more likely to engage in integrative processing early during sentence comprehension when the sentence contains information relevant to the reading goal, provided that the information is familiar to the reader. On the other hand, when the text contents are unfamiliar to the reader, the reading goal influences integrative processing primarily only after

the whole sentence is read through.

Time course of the perspective effects during the second reading of the text

As noted above, participants were asked to read the same texts twice. The purpose of this manipulation was to see whether participants could make more extensive use of the reading goal after constructing a mental representation of the text contents during the initial reading of the text. We also expected the effects of reading goal to appear earlier in the processing time line, particularly for the familiar diseases text. In other words, we predicted the perspective effect to grow bigger and manifest earlier during the second reading compared to the first reading. In order to examine repetition effects, we subtracted for each measure the mean value obtained for the second reading from the value obtained for the first reading.

The results proved our prediction wrong. In all

measures, the size of the perspective effect did not grow bigger or manifest earlier during the second reading compared to the first reading. Naturally, participants read the text much faster during the second reading, but they used the reading goal in the same way as during the first reading of the text. The only significant repetition effect observed in gaze duration and the probability of regression was that of sentence position. Repetition speeded up gaze durations and decreased regressions particularly at the sentence end. This is taken as evidence to suggest that repetition facilitates integrative wrap-up processing done at the sentence end.

To conclude the section on the effects of reading goal, the results presented above suggest that both the initial encoding of words as well as the later integration stage of wrapping up the sentence meaning are influenced by the reading goal, in the present context the assigned reading perspective. Perhaps the most interesting and newsworthy finding is that reading goal exerts its effect early on during processing, especially when reading a text of familiar contents. The early effects were observed both in saccadic programming (i.e., the effect in word skipping) and as modulation of fixation times (i.e., the effect in gaze duration). In the next section we summarize results demonstrating early effects of reading task on eye guidance.

Effects of reading task on eye guidance during reading

Kaakinen and Hyönä (2010) studied effects of reading task on eye movements by comparing proofreading to reading for comprehension. In reading for comprehension the focus is on meaning extraction while in proofreading the emphasis is on sublexical orthographic processing, that is, on searching typing errors in words. To date, the majority of eye movement research has investigated reading for comprehension, but there are also a handful of studies examining eye movements during proofreading.

Daneman, Reingold and Davidson (1995) examined how homophone errors influence eye movements

during proofreading and compared these results to an earlier study (Daneman & Reingold, 1993) in which participants read the same materials with instructions to read for comprehension. They found that gaze durations and total fixation times on target words were longer during proofreading than during reading for comprehension. Wotschak (2009) reports an experiment in which she compared proofreading to reading for comprehension. In line with Daneman et al. (1995), she found that proofreading instructions increased the probability of making more than one fixation per word, which in turn increased gaze durations and total fixation times. Moreover, word skipping probability was lower and the proportion of regressions was higher during proofreading than during reading for comprehension.

Also spatial aspects of eye movements may be influenced by proofreading instructions. Heller (1982) examined the execution of corrective saccades made after return sweeps (i.e., long saccades taking the eyes from the end of one line to the beginning of next line) during proofreading and reading for comprehension. The number of corrective saccades increased in proofreading in comparison to reading for comprehension, suggesting that the optimal point of fixation was more difficult to reach during proofreading. This in turn suggests that the effective visual field may be smaller during proofreading than reading for comprehension. A similar conclusion was reached by Osaka, Osaka and Tsuji (1995) in a moving window study. Also Wotschak (2009) reports results that are in line with this conclusion: for example, saccade amplitudes were shorter and lag and successor effects (i.e., the influence of the preceding and succeeding word on the processing of the target word) were significantly weaker during proofreading than during reading for comprehension. Presumably, a smaller attentional span in proofreading is due to an emphasis on the processing of letter clusters rather than word meanings.

We (Kaakinen & Hyönä, 2010) compared eye movement patterns during proofreading and reading for comprehension by manipulating the length and

frequency of target words. If proofreaders indeed strongly attend to letters and letter clusters, the effect of word length should be stronger in proofreading compared to reading for comprehension. Moreover, on the assumption that proofreaders' effective visual field is relatively narrow, the saccade amplitudes during proofreading should be smaller than during reading for comprehension. On the other hand, if the focus in proofreading is on sublexical processing, effects of word frequency should be diminished in comparison to reading for comprehension. Yet, it is also possible that the word frequency effect would be greater in proofreading. This is because proofreading also entails a post-lexical checking procedure (O'Connor & Forster, 1981) during which the processed letter combination is checked against existing lexical representations to determine whether or not the letter cluster is correctly spelled.

In the experiment, we asked participants first to read a set of sentence for comprehension, which was followed by a proofreading block where they were asked to search for typos in a separate set of sentences. It is important to note that none of the target sentences contained any typographical errors, which always appeared in the filler sentences. Thus, the obtained results are not contaminated by spotting an actual typo. The writing errors were letter transposition errors, where two word-internal letters had swapped position (e.g., Keith→Kieth). We compared readers' eye movements during proofreading and reading for comprehension by performing two sets of analyses, one for the entire target sentences and another for the predetermined target words (the target sentences were adopted from the study of Bertram and Hyönä, 2003). The target words were two-noun compound words, which varied in length (7–9 letters vs. 12–14 letters) and frequency (average frequency of the high-frequency words was 23 occurrences per million words and that of the low-frequency words 2.4 occurrences per million words).

In the sentence-level analyses we compared the mean fixation duration, the mean number of fixations and the mean saccadic amplitude between the two

reading tasks. The results were straightforward. Proofreaders proceeded in the text with longer fixation durations (229 vs. 211 ms), with more fixations (20.6 vs. 16.3) and with shorter saccades (6.9 vs. 8.6 char.) than readers reading the sentences for comprehension.

A similar pattern of results was also apparent in the analysis of the target words. In the word-level analysis the finding of shorter saccades during proofreading than reading for meaning translated into the initial fixation location on target words being significantly closer to the word beginning during proofreading. This finding is presented in Figure 5, which also shows that saccades were launched further into the word for long than short words. The latter finding reflects the preferred landing position effect (Rayner, 1979): readers opt to land their initial fixation toward the word center. The duration of first fixation (see Figure 6) was also affected by reading task. First fixations were significantly longer during proofreading than reading for meaning. They were also longer for low-frequency than high-frequency words and for short than long words. The latter finding reflects the fact that short words are more often read with a single fixation than long words; single fixations are longer than first fixations followed by a refixation on the word (e.g., Hyönä, 1995).

Finally, gaze duration was also influenced by reading task: Gaze durations on target words were longer during proofreading than reading for meaning (see Figure 7). Moreover, the main effects of length and frequency were modified by an interaction with reading task. As predicted, the word length effect was greater in proofreading than reading for meaning. Also the frequency effect was strengthened in proofreading. Finally, task and frequency interacted with each other, suggesting that the frequency effect was greater in proofreading.

To sum up the section on the effects of reading task, it is concluded that (a) task effects appeared very early in the processing time line (i.e., in the duration and location of the initial fixation on words), (b) both temporal and spatial aspects of eye movements were affected by reading task, and (c)

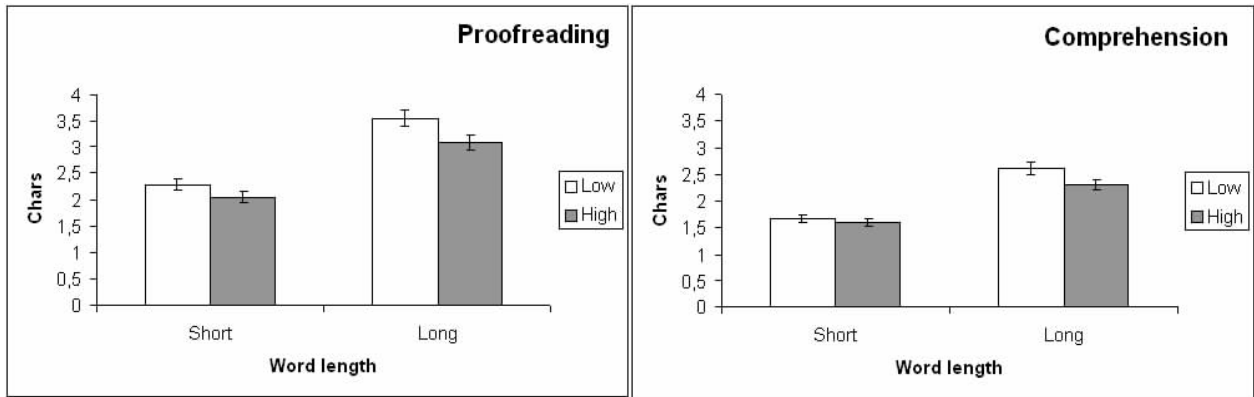


Figure 5. Effect of word length and frequency in the initial landing position on target words during proofreading and reading for meaning.

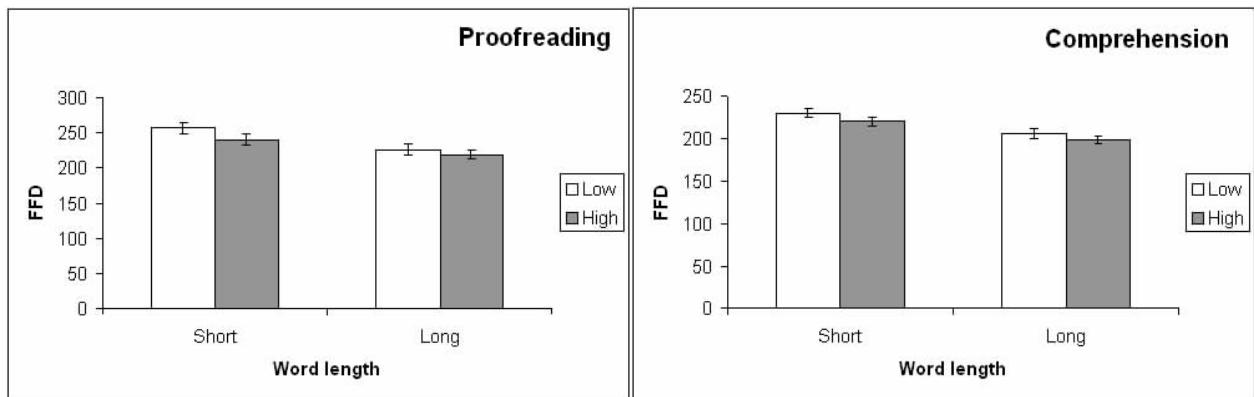


Figure 6. Effect of word length and frequency in first fixation duration on target words during proofreading and reading for meaning.

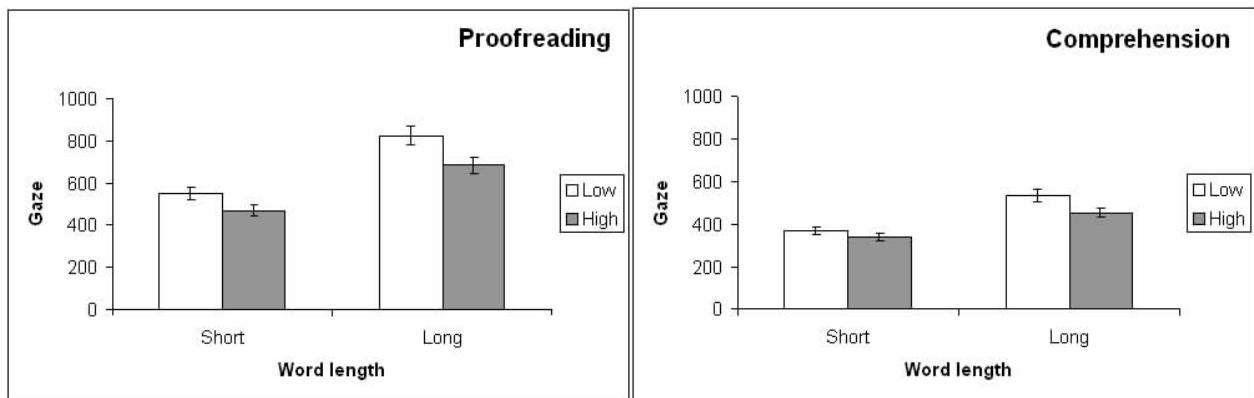


Figure 7. Effect of word length and frequency in gaze duration on target words during proofreading and reading for meaning.

task effects emerged both in local (word-level) and global (sentence-level) eye movement indices. Thus, these data demonstrate that the reading task significantly influences eye guidance during reading. The results are also in line with the view that in proof-

reading the emphasis is on orthographic processing. Moreover, the data are compatible with the view that the attentional span is smaller in proofreading than reading for meaning. Finally, the stronger frequency effect in proofreading suggests that orthographic

checking needed to find typos in words takes place after lexical access.

Conclusions

The two studies (Kaakinen & Hyönä, 2007, 2010) reviewed above demonstrate that a specific reading goal or a reading task exerts pervasive effects on eye guidance during reading. The fact that the goal and task effects were observed not only in local but also global eye movement measures and both spatial and temporal aspects of eye movement control suggests that readers make a global adjustment to eye behavior on the basis of reading task or goal. These data may be explained by gain control models that were proposed during the earlier days of eye movement research (for a review, see Rayner, 1978). Gain control models suggest that, for example, readers may respond to text difficulty by turning down “the gain knob”, which slows down the information intake, which in turn is seen in shorter saccades and longer fixations. The present results demonstrate that readers behave like that in accommodating their reading behavior according to reading task or goal.

Finally, the results are compatible with the view that readers’ attentional span is modulated by reading task. However, as we did not have any direct measure of the attentional span, this conclusion should be considered speculative at this point. Yet, we consider it likely that the global adjustments made to different eye movement parameters are accompanied with (or even mediated by) corresponding fluctuations in attentional span.

References

- Bertram, R., & Hyönä, J. (2003). The length of a complex word modifies the role of morphological structure: Evidence from eye movements when reading short and long Finnish compounds. *Journal of Memory and Language*, 48, 615–634.
- Daneman, M., & Reingold, E. M. (1993). What eye fixations tell us about phonological recoding during reading. *Canadian Journal of Experimental Psychology*, 47, 153–178.
- Daneman, M., Reingold, E. M., & Davidson, M. (1995). Time course of phonological activation during reading: Evidence from eye fixations. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 21, 884–898.
- Heller, D. (1982). Eye movements in reading. In R. Groner & P. Fraise (Eds.), *Cognition and eye movements* (pp. 139–154). Amsterdam: North Holland.
- Hendriks, A. W. (1996). Vergence of eye movements during fixations in reading. *Acta Psychologica*, 92, 131–151.
- Hyönä, J. (1995). Do irregular letter combinations attract readers’ attention? Evidence from fixation locations in words. *Journal of Experimental Psychology: Human Perception and Performance*, 21, 68–81.
- Just, M. A., & Carpenter, P. A. (1980). A theory of reading: From eye fixations to comprehension. *Psychological Review*, 87, 329–354.
- Kaakinen, J. K., & Hyönä, J. (2010). Task effects on eye movements during reading. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 36, 1561–1566.
- Kaakinen, J. K., Hyönä, J., & Keenan, J. (2002). Perspective effects on on-line text processing. *Discourse Processes*, 33, 159–173.
- Kaakinen, J. K., Hyönä, J., & Keenan, J. M. (2003). How prior knowledge, working memory capacity, and relevance of information affect eye-fixations in expository text. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 29, 447–457.
- Kaakinen, J. K., & Hyönä, J. (2005). Perspective effects on expository text comprehension: Evidence from think-aloud protocols, eyetracking and recall. *Discourse Processes*, 40, 239–257.
- Kaakinen, J. K., & Hyönä, J. (2007). Perspective effects in repeated reading: An eyetracking study. *Memory & Cognition*, 35, 1323–1336.
- Kaakinen, J. K., & Hyönä, J. (2008). Perspective-driven text comprehension. *Applied Cognitive Psychology*, 22, 319–334.
- Laycock, F. (1955). Significant characteristics of college students with varying flexibility in reading rate: I. Eye movements in reading prose. *Journal of Experimental Education*, 23, 311–319.
- Lorch, R. F., Jr., Lorch, E. P., & Klusewitz, M. A. (1993). College students’ conditional knowledge about reading. *Journal of Educational Psychology*, 85, 239–252.
- Narvaez, D., van den Broek, P., & Ruiz, A. B. (1999). The influence of reading purpose on inference generation and comprehension in reading. *Journal of Educational Psychology*, 91, 488–496.
- O’Connor, R. E., & Forster, K. I. (1981). Criterion bias and search sequence bias in word recognition. *Memory & Cognition*, 9, 78–92.
- Osaka, N., Osaka, M., & Tsuji, H. (1995). Effective visual field size necessary for proofreading during Japanese text editing. In J. M. Findlay, R. Walker & R. W. Kentridge (Eds.), *Eye movement research: Mechanisms, processes and applications* (pp. 453–463).

- Amsterdam: Elsevier Science.
- Radach, R. Huestegge, L., & Reilly, R. (2008). The role of global top-down factors in local eye-movement control in reading. *Psychological Research*, 72, 675-688.
- Rayner, K. (1978). Eye movements in reading and information processing. *Psychological Bulletin*, 85, 618-660.
- Rayner, K. (1979). Eye guidance in reading: Fixation locations in words. *Perception*, 8, 21-30.
- Rayner, K., Kambe, G., & Duffy, S. A. (2000). The effect of clause wrap-up on eye movements during reading. *Quarterly Journal of Experimental Psychology*, 53A, 1061-1080.
- Rothkopf, E. Z., & Billington, M. J. (1979). Goal-guided learning from text: Inferring a descriptive processing model from inspection times and eye movements. *Journal of Educational Psychology*, 71, 310-327.
- Shebilske, W. L., & Fisher, D. F. (1983). Eye movements and context effects during reading of extended discourse. In K. Rayner (Ed.), *Eye movements in reading: Perceptual and language processes* (pp. 157-179). NY: Academic Press.
- van den Broek, P., Lorch, R. F. Jr., Linderholm, T., & Gustafson, M. (2001). The effects of readers' goals on inference generation and memory for texts. *Memory & Cognition*, 29, 1081-1087.
- Wotschack, C. (2009). Eye movements in reading strategies. (Doctoral dissertation, Universität Potsdam, 2009). *Spektrum Patholinguistik-Schriften*, 1.

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为了落实《国家中长期人才发展规划纲要》和《国家中长期教育改革和发展规划纲要》的要求，贯彻中国共产党十七届五中全会提出的“促进经济长期平稳较快发展和社会和谐稳定”精神，教育部人文社会科学重点研究基地主任、天津师范大学心理与行为研究院院长、“中国心理学会终身成就奖”获得者沈德立教授，发起并组织了全国心理学界著名专家到天津共议“十二五”期间中国心理学界该如何积极行动起来，为社会和谐稳定、人才培养和教育改革服务及应承担的任务。

2010年12月16日，“心理和谐与人才强国”高层论坛暨天津师范大学心理与行为研究院成立10周年大会在天津举行。中国科学院院士陈霖教授；国务院学位委员会心理学学科评议组召集人董奇教授与莫雷教授及成员沈模卫教授、周晓林教授、梁宁建教授、李红教授、白学军教授；中国心理学会现任理事长林崇德教授、候任理事长杨玉芳教授、副理事长乐国安教授、叶浩生教授、王登峰教授、李其维教授、刘华山教授、游旭群教授、张文新教授、秘书长傅小兰教授；以及国家重点学科和国家重点实验室负责人；教育部人文社会科学重点研究基地和教育部重点实验室负责人；全国各心理学博士点带头人；中国心理学刊物主编；长江学者等40多位专家齐聚临会。

教育部社会科学司杨光司长，天津市教委林炎生副主任，天津师范大学党委王璟书记和校长高玉葆教授出席本次论坛，并分别致辞。

高层论坛围绕“心理和谐与人才强国”主题，进行了四方面的研讨：（1）心理学如何为国家经济建设和社会发展服务；（2）“十二五”期间中国心理学的发展；（3）社会和谐稳定的心理基础；（4）高层次人才成长和培养的心理规律。论坛上，每位与会专家都作了精辟阐述。

与会专家一致认为，中国心理学工作者，应该肩负起促进国家经济长期平稳较快发展和社会和谐稳定的责任与使命，在“十二五”期间，要抓住机遇，接受挑战，始终坚持：（1）面向国家发展战略需要；（2）面向世界科学前沿，加强科学问题导向，注重研究成果的理论创新；（3）面向社会发展需要，以心理健康为学科发展重点，前瞻性部署灾害与创伤心理和网络行为研究，促进全民心理健康水平提高，优化人民群众的生活品质；（4）注重心理学在高新技术、国防安全、航空航天、教育改革、创新人才培养、人力资源开发等方面的应用；（5）在和谐社会建设中，应该重视心理和谐的作用，积极建议国家在构建国家经济社会发展指标时要考虑人文指标。

这次论坛上各位专家的发言将整理后印刷成册，发给各心理学单位。

《心理和谐与人才强国》高层论坛秘书处

2010.12